Jita - Trader High level algorithm overview:

JJB

1. Goal:

- Maximize ISK gain per trip while simultaneously optimizing for minimum # of jumps & AU travelled

2. Inputs

- global market data
- all region's buy + sell orders
- Current location/stats of Hauling Ship
- current on-hand capital of Hauling Ship

KEY:

= initial decomp

= comments after short break

= comments after long break

= Comments after Final read through

3. General process:

from step 4

Step 1: get Current info
- pull global & current region market data
- pull on-hand capital
- pull ship stats

Step 2: determine initial buy
- Filter global & region market data to only include items that will fit into hauler, & can be purchased w/ on-hand capital
- find lowest priced item compared to global average & Estimate Initial ISK investment
  - ↳ optimize buy to be within ? jumps of hauler's current location

*for starters choose lowest relative to global average, change later to search all regions for greatest price Δ

Step 3: find (mostly) optimized sell
- poll all regions to find highest sell price compared to global average
- Save top ? regions, & check for other buy offers significantly lower than global average
- weight regions according to sell price, distance from player, & estimated profit of next trade
- Set waypoint to top sell option station, undock ship, activate autopilot

This reward function is going to be an absolute trainwreck, we'll need to revise this constantly, & figure out a simple-ish next profit prediction system

↳ maybe look into storing items w/ high price Δ's as the market updates?

↳ does global average hold up as a method of determining a good buy?

↳ from next day Jackson: maybe? It's comically easy to check, but a really low buy price has no guarantee of an efficient or profitable sell

for starters we should only fulfill other's buy orders, but we should look at implementing some sort of ML that gets fed all market data & determines if listing a sell order is a better alternative

Step 4: Execute sell
- Dock at Station and move goods to Station Storage
- open market tab & perform sell order
- pull wallet transactions to determine exact profit & save relevant metrics (total profit, profit per jump, profit per volume, etc.)

probably going to use these to verify the performance of the algorithm/system

to step 1

General algorithm decision tree

Is the item cheaper near me than in another region?

- No → find a different good.
- Yes → Where is the item more expensive than we bought it?
  - far → Is the profit margin high enough to justify the distance?
    - No → find a different good
    - Yes → Does the target region also have a comparatively cheap good to trade?
  - near-ish → Does the target region also have a comparatively cheap good to trade?
    - Yes → Seems great, set the autopilot & go.
    - No → Is the profit margin high enough to justify the further distance to next good?
      - Yes → Ship it.
      - No → Find a different good